UNIT 3 – Specialized Testing: API & Database Testing

Exhaustive Study Notes covering REST Architecture, Postman, SQL Queries, Joins & Data Integrity

1. API Testing Fundamentals

Application Programming Interface (API) testing validates backend logic, performance, security, and data reliability without invoking the graphical user interface (GUI).

Advantages of API Testing: • Early Testing: Test business logic before UI is developed. • Fast Execution: API calls execute in milliseconds compared to slow UI rendering. • Language Independent: Data exchanged via standardized JSON or XML payloads.

2. REST API & HTTP Verbs

REpresentational State Transfer (REST) is an architectural style for web services.

Standard HTTP Verbs: • GET: Retrieve data from server (Read-only, Idempotent). • POST: Create a new resource on server. • PUT: Update/Replace entire existing resource. • PATCH: Partially update specific fields of a resource. • DELETE: Remove a resource from server.

Client - REST API - Database Architecture & HTTP Methods

Diagram: *Client - REST API - Database Architecture & HTTP Verbs.*

3. JSON Payload & Format

JavaScript Object Notation (JSON) is a lightweight format used for API payload data exchange.

```
{
  "student_id": 101,
  "name": "Asha Sharma",
  "email": "asha@example.com",
  "is_active": true,
  "courses": ["Django", "Software Testing"],
  "address": {
    "city": "Mumbai",
    "pincode": "400001"
  }
}
```

4. HTTP Response Status Codes

Status Code Taxonomy: • 2xx Success: 200 OK, 201 Created, 204 No Content. • 3xx Redirection: 301 Moved Permanently, 302 Found. • 4xx Client Error: 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found. • 5xx Server Error: 500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable.

5. Postman API Testing Tool

Postman is a popular API testing tool allowing testers to organize HTTP requests into Collections, pass Environment Variables, and execute JavaScript assertions.

```
// Postman Test Assertions Script (Tests Tab)
pm.test("Status code is 200 OK", function () {
    pm.response.to.have.status(200);
});

pm.test("Response contains user email", function () {
    var jsonData = pm.response.json();
    pm.expect(jsonData.email).to.eql("asha@example.com");
});
```

Postman API Test Execution & Assertion Pipeline

Diagram: *Postman API Test Execution & Assertion Pipeline.*

6. Database Testing & SQL Basics

Database Testing validates backend data storage, schema compliance, field data types, foreign key constraints, triggers, and stored procedures.

DDL vs DML: • Data Definition Language (DDL): `CREATE`, `ALTER`, `DROP` (Schema structure). • Data Manipulation Language (DML): `SELECT`, `INSERT`, `UPDATE`, `DELETE` (Data content).

7. SQL Queries - CRUD Operations

Essential SQL Commands for QA Engineers:

```
-- READ Data
SELECT id, name, email FROM students WHERE age >= 18 ORDER BY name ASC;

-- CREATE Data
INSERT INTO students (name, email, age) VALUES ('Rohan Verma', 'rohan@example.com', 22);

-- UPDATE Data
UPDATE students SET email = 'rohan_new@example.com' WHERE id = 105;

-- DELETE Data
DELETE FROM students WHERE id = 105;
```

8. SQL Joins & Data Integrity

Joins combine records from two or more tables based on a related foreign key column: • INNER JOIN: Returns matching rows in both tables. • LEFT JOIN: Returns all rows from left table and matching rows from right table. • RIGHT JOIN: Returns all rows from right table and matching rows from left table. • FULL OUTER JOIN: Returns all records when a match exists in left or right table.

ACID Properties Verification: Atomicity, Consistency, Isolation, Durability.

```
-- Sample SQL Join Query for Validation
SELECT s.name, d.department_name
FROM students s
INNER JOIN departments d ON s.department_id = d.id;
```

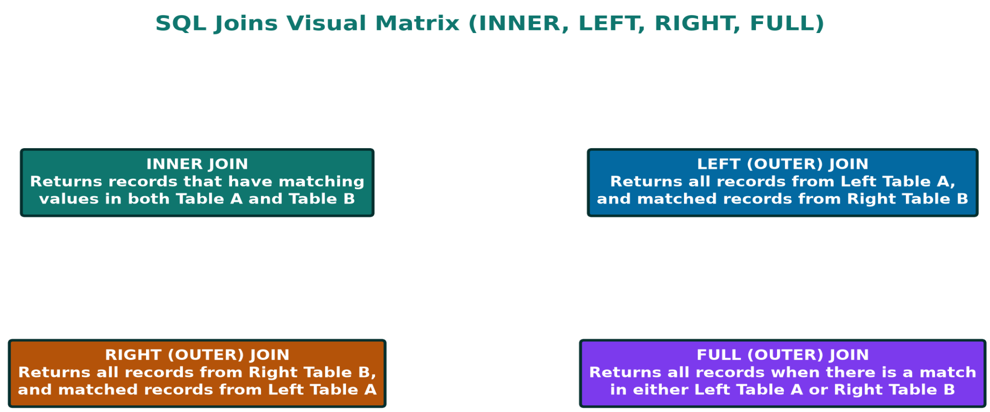

Diagram: *SQL Joins Visual Matrix (INNER, LEFT, RIGHT, FULL).*